RGIPT

Subject: AI

Class: 12

Time Allowed: 45 minutes **Maximum Marks: 10**

All questions are compulsory unless stated otherwise.

Name: ___________________________

Roll Number: ____________________

Class: 12 Section: ________

Section A: Multiple Choice Questions

Attempt all questions. Each question carries 1 mark.

1. [Easy] What is the primary purpose of an activation function in a neural network? [1 Marks]

2. [Easy] Which loss function is most commonly used for regression tasks? [1 Marks]

3. [Easy] In gradient descent, what does the learning rate control? [1 Marks]

4. [Easy] Which activation function outputs max(0, x)? [1 Marks]

Section B: Short Answer Questions

Answer briefly. Each question carries 2 marks.

1. [Moderate] Explain why stacking multiple linear layers without activation functions is equivalent to a single linear layer. [2 Marks]

2. [Moderate] List and describe the four steps of the training loop in neural networks. [2 Marks]

3. [Challenging] Given the linear relationship $200 per sq ft, the actual price for a 2,500 sq ft house is $500,000. If a model predicts $350,000 for this house, compute the Mean Squared Error (MSE). [2 Marks]

End of Question Paper

Answer Key:

1. To introduce non-linearity so the network can model complex functions.

2. Mean Squared Error (MSE).

3. The size of the step taken toward the minimum loss.

4. ReLU (Rectified Linear Unit).

5. Each linear layer computes $y = W x + b$. Composing them yields $y = (W_2W_1) x + (W_2b_1 + b_2)$, which is again a linear transformation. Thus multiple linear layers collapse into one overall linear layer, providing no extra modelling power.

6. 1) Forward Pass – inputs flow through the network to produce a prediction. 2) Calculate Loss – compare prediction with true target to obtain a scalar error. 3) Backpropagation – propagate the error backward to compute gradients for each weight. 4) Update Weights – adjust weights using gradient descent (or a variant) to reduce loss.

7. Error = 350,000 − 500,000 = −150,000; Squared error = $(−150,000)^2$ = 22,500,000,000. With a single example (n=1), MSE = 22,500,000,000.